

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech**

Semester: VIII

Name of the Course: Multimedia Systems and Data Compression

Course Code: TCS 823

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	Define multimedia presentation and explain its key components. Explain the need for digital representation of data in brief.	
	OR	
(b)	Explain the relationship between sampling rate and bit depth in digital media.	
Q2	(10 Marks)	CO1
(a)	Discuss the process of Analog to Digital Conversion (ADC) and Digital to Analog Conversion (DAC) with suitable examples.	
	OR	
(b)	Describe the role of Nyquist's Theorem in analog to digital conversion with the help of suitable example.	
Q3	(10 Marks)	CO1 ,CO 2
(a)	What is Pulse Modulation? Describe its different types and their applications.	
	OR	
(b)	Explain SMIL and its role in multimedia presentations. Provide an example of its usage.	
Q4	(10 Marks)	CO1 ,CO 2
(a)	What is Run-Length Encoding (RLE)? Differentiate between lossless and lossy data compression with examples.	
	OR	
(b)	Explain Shannon-Fano coding with an example. How does it differ from Huffman coding?	
Q5	(10 Marks)	CO1
(a)	Compare Huffman Coding, Arithmetic Coding, and Dictionary-based compression methods. Discuss their advantages and limitations.	
	OR	
(b)	You are designing a multimedia presentation for an educational institution. How will you integrate different multimedia elements to make the presentation more engaging? Provide a step-by-step approach.	CO2

